Industrial Safety Processes

Hazard Assessments

Heart Hat SAFETY

Content

- Hazards & Assessment
- Controls
- Formal Hazard Assessment Process

Heart Hat
SAFETY

Hazard ID, Assessment & Control

- The function of an occupational health and safety (OHS) management system is to:
  - Identify hazards
  - Assess hazards
  - Control hazards

Heart Hat
SAFETY

Hazard

"A situation, condition or thing that may be dangerous to the safety or health of workers."

Heart Hat SAFETY

Jobs, Skills, Training and Labour, Government of Alberta, *Hazard Assessment and Control: A Handbook for Alberta Employers and Workers*. Edmonton, Alberta (2015)., p. 1.

Things, Etc.

Hazard	Thing / Situation / Condition
Ice on sidewalk	Thing
Wildlife	Thing
Faulty equipment	Condition / Thing
Working at heights	Situation
Fatigue	Situation / Condition
Falling object	Thing
Confined space	Thing

Heart Hat
SAFETY

Health & Safety Hazards

- There are 2 main types of hazards:
  - Health hazards
  - Safety hazards
- Those familiar with Certificate of Recognition audits will be aware that both types of hazards are required for formal hazard assessments
- Some hazard assessments indicate if a hazard is a “health” or a “safety” hazard, but this is not necessary

Heart Hat
SAFETY

Health Hazards

- "A health hazard is anything that could harm someone's health, either immediately or overtime." (p. 12)
- Health hazards can sometimes be hard to spot (see section on hazard categories below) and identify
- Increasingly regulators, the public, unions and "older" organizations are concerned about health hazards in the workplace

Heart Hat SAFETY

Jobs, Skills, Training and Labour, Government of Alberta, *Hazard Assessment and Control: A Handbook for Alberta Employers and Workers*. Edmonton, Alberta (2015)., p. 12.

Safety Hazards

- "A safety hazard is anything that could cause injury or damage." (p. 12)
- Most of what is seen as a "hazard" in a workplace is a "safety" hazard
- These are easier to identify, as we will see in the categories of hazards in a little bit

Heart Hat SAFETY

Jobs, Skills, Training and Labour, Government of Alberta, *Hazard Assessment and Control: A Handbook for Alberta Employers and Workers*. Edmonton, Alberta (2015)., p. 12.

Things, Etc.

Health Hazards	Safety Hazards
Vapours & Fumes	Sharp Objects
Bodily Fluids	Wildlife
Asbestos	Confined Space
Ultraviolet Radiation	Working at Height
Bullying	Extreme Temperatures
Carcinogenic Chemical	Glare / Bright Lights
Fatigue	Low Light / Darkness

Heart Hat
SAFETY

Some Hazards Could be Both Types

- Violence
  - Physical harm (safety), psychological harm (health)
- Radiation
  - Physical harm (safety), illness / disease (health)
- Major physical injury
  - Physical harm (safety), complications over time (health)
- Safety course
  - Brain damage (safety), develop personality disorder (health)

Heart Hat
SAFETY

4 Categories of Hazards

1. Physical
2. Chemical
3. Biological
4. Psychological

Heart Hat
SAFETY

Examples of Hazards by Category

Physical	Chemical	Biological	Psychological
Noise	Cleaning Chemicals	Viruses & Bacteria	Harassment
Vibration	Fuels	Animal / Human Waste	Bullying
Radiation	Gases	Bodily Fluids	Violence
Extreme Temperatures	Vapours / Fumes	Moulds	Fatigue

Heart Hat
SAFETY

Assessing Hazards

- This is done to prioritize which hazards are addressed first
- You want to pay more time, attention and effort to address more serious threats to health and safety
- Individual hazards are assessed as the risk they pose may vary based on tasks or circumstances
  - Fatigue in office vs. fatigue while driving

Heart Hat
SAFETY

Multiple Hazard Control Hierarchies

- Many different hazard control hierarchies are commonly used, but few are legislated
- They all following the same priority of first elimination, then proceed down the list
- The key thing is to use one and stick with it
- Here we will explain the hierarchy in Alberta legislation

Heart Hat
SAFETY

Alberta Hierarchy of Hazard Controls

1. Elimination
2. Engineering Controls
3. Administrative Controls
4. Personal Protective Equipment
5. Combination of controls 2 to 4.

Heart Hat
SAFETY

Elimination

- Must be considered first
- This is removing the hazard from the workplace – the hazard does not exist in your workplace

Heart Hat
SAFETY

Engineering Controls

- If you can eliminate a hazard, engineering controls must be considered first
- In the “hazard path” engineering controls should be placed as close as possible to the hazard
- Normally engineering controls “work” without intervention from the worker…
- But they may need to be activated – i.e., turn on lights, turn on ventilation fan

Heart Hat
SAFETY

Engineering Controls

- Substitution – if elimination is not practical, try substituting or replacing one substance or process with another. Examples include:
  - substituting a safer substance for a more hazardous one
  - replacing hazardous operations with less hazardous operations

Heart Hat SAFETY

Government of Alberta, Occupational Health and Safety Code 2009 Explanation Guide (2009), p. 2-17

Administrative Controls

- These are controls that rely on a worker – or somebody – doing something to avoid or control a hazard
- These are the “procedures and practices” of the workplace
- These include anything that requires a worker to interpret and respond to in an appropriate way

Heart Hat
SAFETY

Personal Protective Equipment

- Last line of defense for the worker
- Should be the last control implemented by the employer
- Represents the physical equipment worn by an individual worker to protect against a hazard
- PPE is intended only to reduce the severity of an incident created by a hazard, not reduce the likelihood of an incident
- If people are wearing PPE, there should be higher-order controls in place

Heart Hat
SAFETY

Examples of Hazard Controls

Elimination	Engineering	Administrative	PPE
Remove Hazard from Workplace	Barriers / Guards	Training	Hard Hat
	Ventilation	Signage	N95 Mask
	Automatic Shut Offs	Procedures	Gloves
	Speed Limiting Devices	Hazard Assessments	Steel-toe Boots

Heart Hat
SAFETY

Formal Hazard Assessments

- These are the primary proactive tools for the identification of hazards
- These cover all operations of an organization and all activities performed
- Individual hazards are identified and assessed to determine priority for control
- Commonly known under different names & acronyms
  - JSA, JHA, THA, Task Analysis, SSSP, Site Safety Plan, Hazard Register, etc.

Heart Hat SAFETY

Formal Hazard Assessments

- Typically limited to a specific job / position or a defined task
- Usually the minimal legal requirement to meet OHS due diligence
- Safety by Design would require formal hazard assessments on all activities and operations to be completed, reviewed as a bundle, and connected elements to be assessed and addressed

Heart Hat
SAFETY